Flask入门 - 2019年最新

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1. 安装

1.1 安装 virtualenv

```
pip install virtualenv
```

1.2 创建虚拟环境

```
virtualenv venv
```

1.3 激活虚拟环境

```
source venv/bin/activate
```

1.4 安装 Flask

```
pip install flask
```

2. 最简单的应用

2.1 hello.py

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def index():
    return 'Hello World'

if __name__ == '__main__':
    app.run()
```

2.2 运行

```
python hello.py
```

```
* Serving Flask app "hello"
* Environment: production
  WARNING: Do not use the development server in a production environment.
  Use a production WSGI server instead.
* Debug mode: off
* Running on http://127.0.0.1:5000/ (Press CTRL+C to quit)
```

2.3 访问

```
curl http://127.0.0.1:5000/
```

```
Hello World
```

3. 路由

3.1 hello.py

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def index():
    return 'Hello World'

@app.route('/hello')
def hello():
    return 'Hello'

@app.route('/user/<name>')
def user(name):
    return 'User %s' % name

@app.route('/post/<int:post_id>')
def post(post_id):
    return 'Post %d' % post_id

@app.route('/path/<path>')
def path(path):
    return 'Path %s' % path

@app.route('/login', methods=['GET', 'POST'])
def login():
    return 'Login'

if __name__ == '__main__':
    app.run()
```

3.2 访问

```
curl http://127.0.0.1:5000/
curl http://127.0.0.1:5000/hello
curl http://127.0.0.1:5000/user/tom
curl http://127.0.0.1:5000/post/123
curl http://127.0.0.1:5000/path/a/b/c
curl -X POST http://127.0.0.1:5000/login
```

```
Hello World
Hello
User tom
Post 123
Path a/b/c
Login
```

4. 模板

4.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/user/<name>')
def user(name):
    return render_template('user.html', name=name)

if __name__ == '__main__':
    app.run()
```

4.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>
</body>
</html>
```

4.3 templates/user.html

```
templates/user.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>User</title>
</head>
<body>
<h1>Hello, {{ name }}</h1>
</body>
</html>
```

4.4 访问

```
curl http://127.0.0.1:5000/
curl http://127.0.0.1:5000/user/tom
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>
</body>
</html>
<!DOCTYPE html>
<html>
<head>
<title>User</title>
</head>
<body>
<h1>Hello, tom</h1>
</body>
</html>
```

5. 静态文件

5.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

5.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
<link rel="stylesheet" href="/static/style.css">
<script src="/static/script.js"></script>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

5.3 static/style.css

```
static/style.css
```

```
body {
    color: red;
}
```

5.4 static/script.js

```
static/script.js
```

```
alert('Hello');
```

5.5 static/image.jpg

```
static/image.jpg
```

5.6 访问

```
curl http://127.0.0.1:5000/
curl http://127.0.0.1:5000/static/style.css
curl http://127.0.0.1:5000/static/script.js
curl -O http://127.0.0.1:5000/static/image.jpg
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
<link rel="stylesheet" href="/static/style.css">
<script src="/static/script.js"></script>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
body {
    color: red;
}
alert('Hello');
```

6. 请求和响应

6.1 hello.py

```
from flask import Flask, request, response, redirect, abort, jsonify

app = Flask(__name__)

@app.route('/')
def index():
    return 'Index'

@app.route('/hello')
def hello():
    return 'Hello'

@app.route('/user/<name>')
def user(name):
    return 'User %s' % name

@app.route('/post/<int:post_id>')
def post(post_id):
    return 'Post %d' % post_id

@app.route('/path/<path>')
def path(path):
    return 'Path %s' % path

@app.route('/login', methods=['GET', 'POST'])
def login():
    return 'Login'

if __name__ == '__main__':
    app.run()
```

6.2 访问

```
curl http://127.0.0.1:5000/
curl http://127.0.0.1:5000/hello
curl http://127.0.0.1:5000/user/tom
curl http://127.0.0.1:5000/post/123
curl http://127.0.0.1:5000/path/a/b/c
curl -X POST http://127.0.0.1:5000/login
```

```
Index
Hello
User tom
Post 123
Path a/b/c
Login
```

7. 会话

7.1 hello.py

```
from flask import Flask, session, redirect, request, render_template

app = Flask(__name__)
app.secret_key = 'secret'

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/login', methods=['GET', '']
    return render_template('login.html')
    session['username'] = request.form['username']
    return redirect('/')

@app.route('/logout')
def logout():
    session.pop('username')
    return redirect('/')

if __name__ == '__main__':
    app.run()
```

7.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

7.3 templates/login.html

```
templates/login.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Login</title>
</head>
<body>
<h1>Login Page</h1>

</body>
</html>
```

7.4 访问

```
curl http://127.0.0.1:5000/
curl http://127.0.0.1:5000/login
curl http://127.0.0.1:5000/logout
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
<!DOCTYPE html>
<html>
<head>
<title>Login</title>
</head>
<body>
<h1>Login Page</h1>

</body>
</html>
```

8. 闪现

8.1 hello.py

```
from flask import Flask, flash, redirect, render_template

app = Flask(__name__)
app.secret_key = 'secret'

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/login')
def login():
    flash('Login')
    return redirect('/')

@app.route('/logout')
def logout():
    flash('Logout')
    return redirect('/')

if __name__ == '__main__':
    app.run()
```

8.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

8.3 访问

```
curl http://127.0.0.1:5000/
curl http://127.0.0.1:5000/login
curl http://127.0.0.1:5000/logout
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

9. 蓝图

9.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

9.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

9.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

10. 模板继承

10.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

10.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

10.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

11. 自定义模板过滤器

11.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

11.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

11.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

12. 自定义模板函数

12.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

12.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

12.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

13. 上下文处理器

13.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

13.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

13.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

14. 钩子函数

14.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

14.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

14.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

15. 数据库操作

15.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

15.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

15.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

16. 表单

16.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

16.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

16.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

17. 文件上传

17.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

17.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

17.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

18. 邮件

18.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

18.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

18.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

19. 日志

19.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

19.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

19.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

20. 单元测试

20.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

20.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

20.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

21. 部署

21.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

21.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

21.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

22. 扩展

22.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

22.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

22.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

23. 项目实战

23.1 hello.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run()
```

23.2 templates/index.html

```
templates/index.html
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

23.3 访问

```
curl http://127.0.0.1:5000/
```

```
<!DOCTYPE html>
<html>
<head>
<title>Index</title>
</head>
<body>
<h1>Index Page</h1>

</body>
</html>
```

参考

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